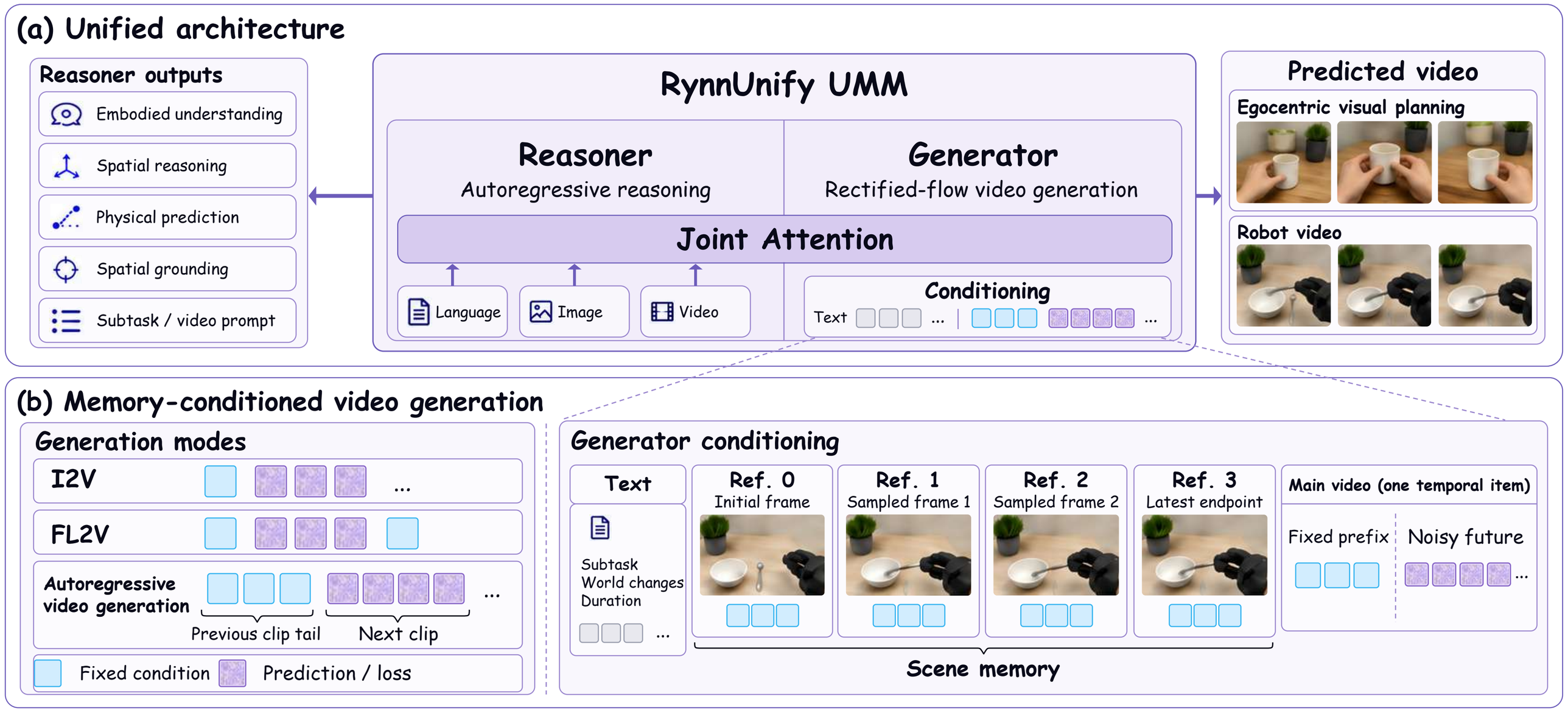

(a) Unified architecture
Reasoner outputs
Embodied understanding
Spatial reasoning
Physical prediction
Spatial grounding
Subtask / video prompt
RynnUnify UMM
Reasoner
Autoregressive reasoning
Generator
Rectified-flow video generation
Joint Attention
Language
Image
Video
Conditioning
Text
Predicted video
Egocentric visual planning
Robot video
(b) Memory-conditioned video generation
Generation modes
I2V
FL2V
Autoregressive video generation
Previous clip tail
Next clip
Fixed condition
Prediction / loss
Generator conditioning
Text
Subtask
World changes
Duration
Ref. 0
Initial frame
Ref. 1
Sampled frame 1
Ref. 2
Sampled frame 2
Ref. 3
Latest endpoint
Scene memory
Main video (one temporal item)
Fixed prefix
Noisy future